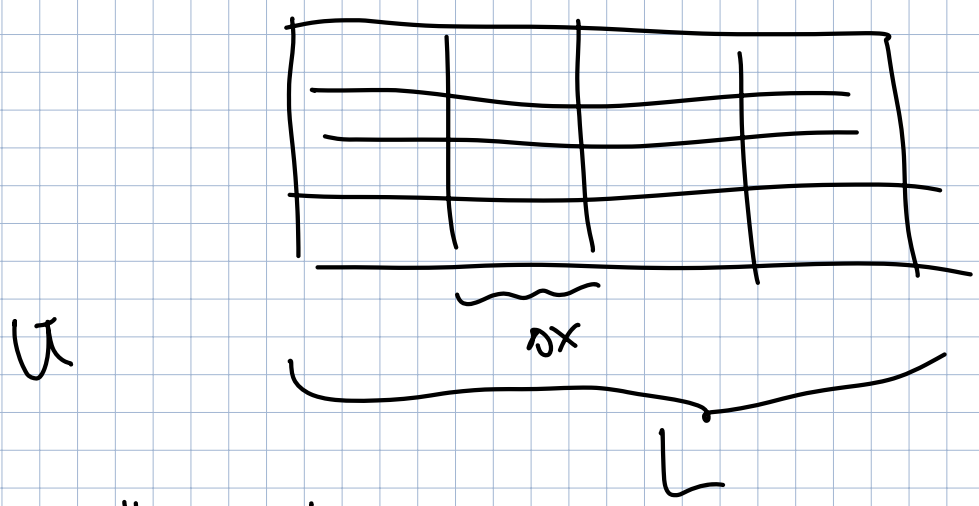
Closures and corrections

velocity

viscosity

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \nu \nabla^2 u_i.$$

DNS – direct numerical sim.



smallest scales:

Kolmogorov length $= \eta = \left(\frac{\nu^3}{\varepsilon}\right)^{1/}$

$$\Delta x = \frac{\eta}{L}$$

or

$$N = \frac{L}{\eta}$$

$$Re = \frac{UL}{\nu}$$

At high Re:

$$\varepsilon \sim \frac{U^3}{L}$$

$$\rightarrow \frac{\eta}{L} \sim Re^{-3/4}$$

or $N \sim Re^{3/4}$

$Re \rightarrow Re \times 10$

$\rightarrow N^3 \times 178$

$$N^3 \sim Re^{9/4}$$

$Re \sim 10^6$

~ 31 trillion cells

Two "fixes":

① RANS

u_i = mean + fluctuation

$= \bar{u}_i + u_i'$

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial x_i} + \nu \nabla^2 \bar{u}_i - \frac{\partial \overline{u_i' u_j'}}{\partial x_j}.$$ — ?

$$\tau_{ij} - \frac{1}{3}\tau_{kk}\delta_{ij} = -2\nu_t \bar{S}_{ij}$$

visc.

$$\bar{S}_{ij} = \frac{1}{2}\left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i}\right)$$

② LES ← sim

large, eddy

take filtered ϕ:

$$\bar{\phi}(\underline{x}, t) = \iint \phi(\underline{r}, t)\, G(\underline{x}-\underline{r}, t-\tau)\, d\tau\, d\underline{r}$$

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial x_i} + \nu \nabla^2 \bar{u}_i - \frac{\partial \tau_{ij}^{sgs}}{\partial x_j}$$

subgrid scale stress

"Closure" in numerics:

$u(x,t) =$ density at x at time t.

$$\int_{x_1}^{x_2} u(x,t)\,dx = \text{total mass in } [x_1, x_2] \text{ at } t.$$

$F_1(t)$ rate of change at x_1

$(F_i > 0$ "to the right")

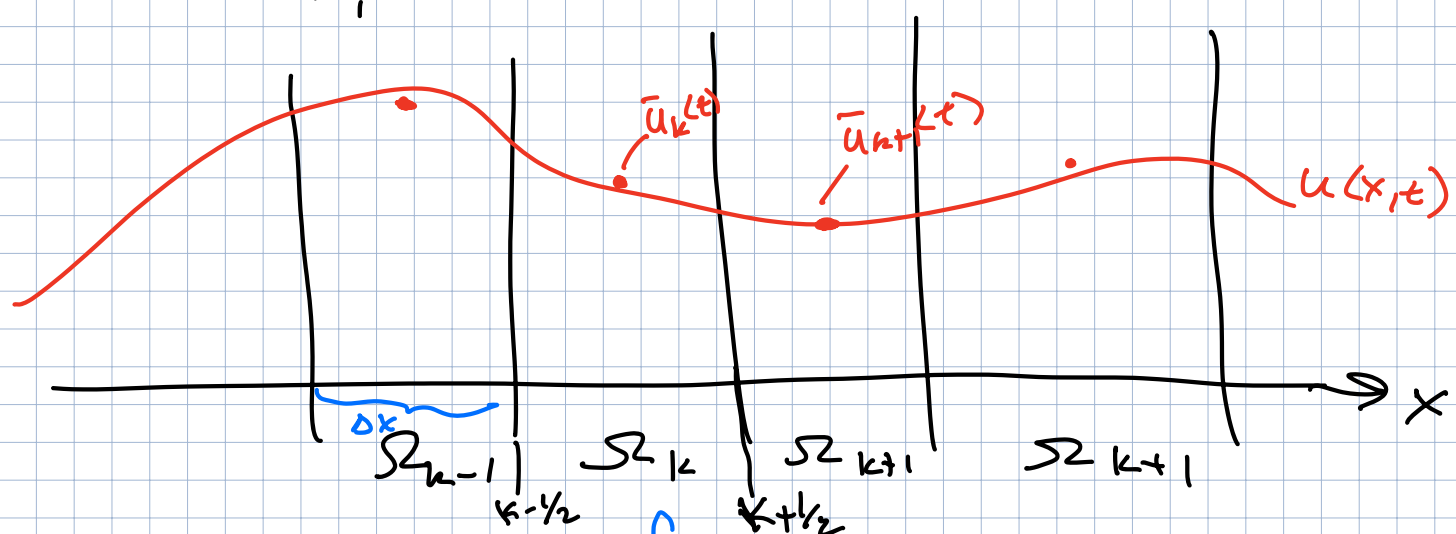
$$\frac{d}{dt}\int_{x_1}^{x_2} u(x,t)\,dx = F_1(t) - F_2(t)$$

$$= f(u(x_1,t)) - f(u(x_2,t))$$

$$= -f(u(x,t))\Big|_{x_1}^{x_2}$$

$$\rightarrow \frac{d}{dt}\int_{x_1}^{x_2} u(x,t)\,dx + \int_{x_1}^{x} \frac{d}{dx} f(u(x,t))\,dx = 0$$

$$\rightarrow \int_{x_1}^{x_2} u_t + (f(u))_x \,dx = 0$$



let $\bar{u}_k(t) = \frac{1}{\Delta x}\int_{\Omega_k} u(x,t)\,dx$

$$\frac{d\,\bar{u}_k(t)}{dt} + \frac{f(u(x_{k+1/2},t)) - f(u(x_{k-1/2},t))}{\Delta x} = 0$$

① use discrete values

u_k = average

② define numerical flux

$f(u) = \frac{u^2}{2}$

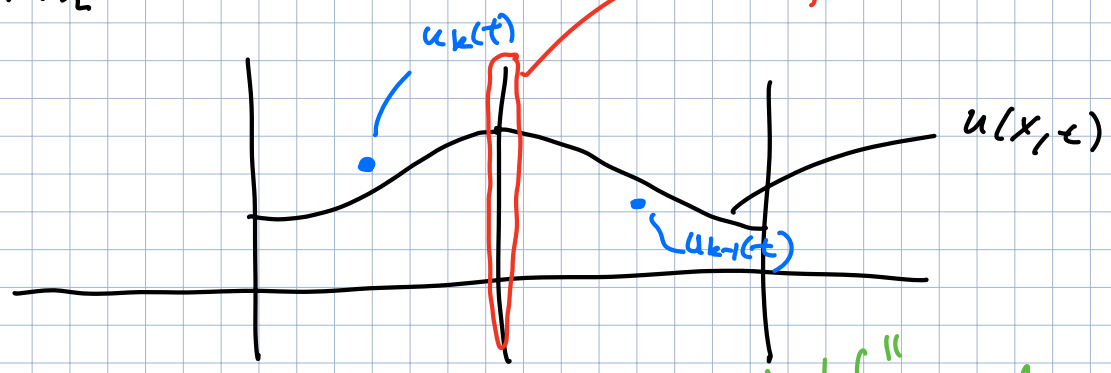
③ ODE solver (Euler)

$$\frac{d\,u_k(t)}{dt} + \frac{\hat{f}_{k+1/2}(t) - \hat{f}_{k-1/2}(t)}{\Delta x} = 0$$

$\hat{}$ ← num. flux

$\hat{f}_{k+1/2}(t) =$

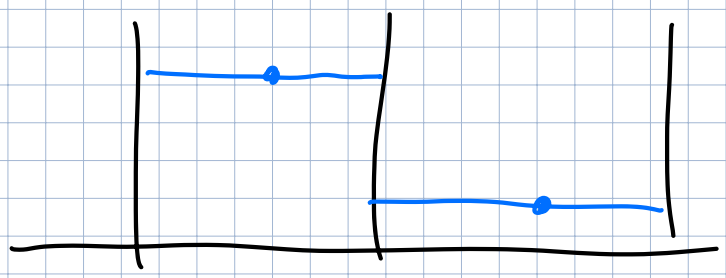
what is the flux here?



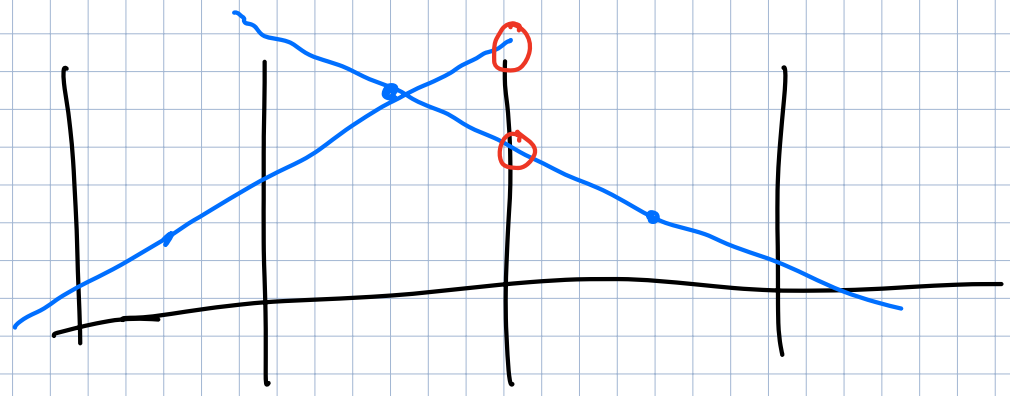
Ω_k $k+1/2$

"from the left" "from the right"

$$= f^*\left(u^-_{k+1/2}(t),\ u^+_{k+1/2}(t)\right)$$



constant



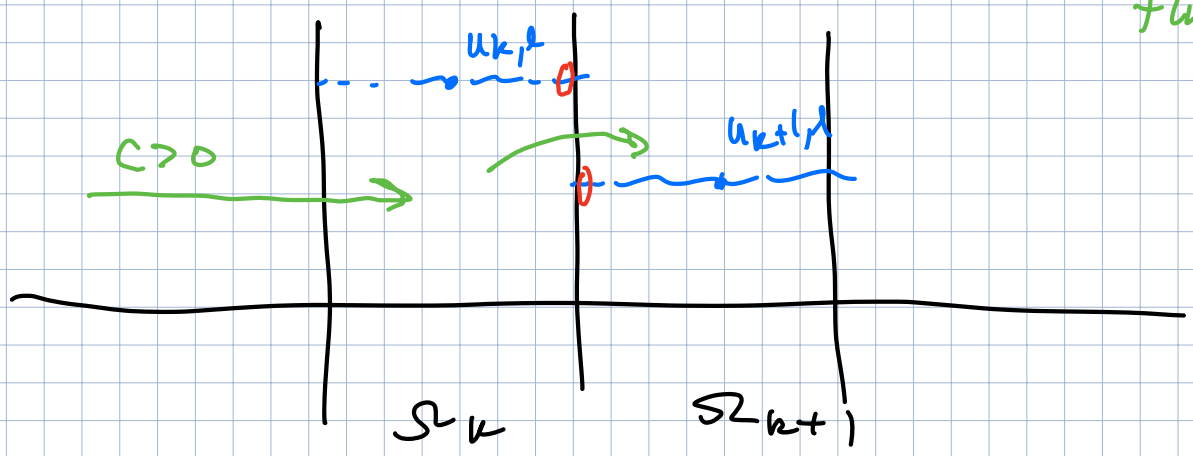
let $u_t + c u_x = 0 \qquad c > 0$

$$f^*_{k+1/2,\,\ell} = f^*(\bar{u}_{k\ell}, \bar{u}_{k+1\ell})$$

time ℓ

$$= \boxed{c\,\bar{u}_{k\ell}}$$

"upwind" flux



$$\frac{\bar{u}_{k\ell+1} - \bar{u}_{k\ell}}{\Delta t} + \frac{c\,\bar{u}_{k\ell} - c\cdot\bar{u}_{k-1\ell}}{\Delta x} = 0$$

drop

$$\rightarrow \boxed{u_{k\ell+1} = u_{k\ell} - c\cdot\frac{\Delta t}{\Delta x}\left(u_{k\ell} - u_{k-1\ell}\right)}$$

ℓ index in time

$$u_{k\ell+1} = u_{k\ell} - c\frac{\Delta t}{\Delta x}\left(u_{k\ell} - u_{k-1\ell}\right)$$

k index in space

advection speed

Take "u"

operator

Want $f = L(u)$

instead we get $\hat{f}$

$\rightarrow \hat{f} - f$ = forward error

Alternative: what operation is $\hat{f}$ exact for

How much error is required in u to account for $\hat{f}$?

$\hat{u} - u$ = backward error

$\Rightarrow L(\hat{u}) = \hat{f}$

$$\begin{tikzcd}
u \arrow[r, "L"] \arrow[dr] & f = L(u) \arrow[d, dotted, "\text{fwd}"] \\
\hat{u} \arrow[u, dotted, "\text{bkwd}"] \arrow[r, "L"'] & \hat{f} = \hat{L}(u) = L(\hat{u})
\end{tikzcd}$$

Example

$f = \sqrt{u}$ let $u = 2$

Suppose we get $\hat{f} = 1.4$

fwd error: $|\hat{f} - f| = |1.4 - 1.4142\ldots|$

$= .0142\ldots$

Find $\hat{u}$ s.t. $\sqrt{\hat{u}} = 1.4$

$\rightarrow \hat{u} = 1.96$

$\rightarrow$ bkd error $= |\hat{u} - u| = .04$

$$\text{condition \#} = \frac{\text{relative forward error}}{\text{" backward "}}$$

$$= \frac{\left|\frac{\hat{f} - f}{f}\right|}{\left|\frac{\hat{u} - u}{u}\right|}$$

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back to $u_t + c u_x = 0$

$$u_{k,l+1} = u_{k,l} - c\frac{\Delta t}{\Delta x}\left(u_{k,l} - u_{k-1,l}\right)$$

cosistent: use exact solution, what is the error.

$$\tau = O(\Delta t, \Delta x)$$

stable: $c\dfrac{\Delta t}{\Delta x} < 1$

$$u_{k\ell+1} = u(x_k, t_{\ell+1})$$

$$= \left[u + u_t \Delta t + u_{tt}\frac{\Delta t^2}{2} + O(\Delta t^3)\right]_{k,\ell}$$

$$u_{k-1\ell} = \left[u - u_x \Delta x + u_{xx}\frac{\Delta x^2}{2} - u_{xxx}\frac{\Delta x^3}{6} + O(\Delta x^4)\right]_{k\ell}$$

$$\cancel{u} + u_t \Delta t + u_{tt}\frac{\Delta t^2}{2} + O(\Delta t^3) = \cancel{u} - \lambda\left(\cancel{u} + \cancel{-u} + u_x \Delta x - u_{xx}\frac{\Delta x^2}{2} + u_{xxx}\frac{\Delta x^3}{6} - O(\Delta x^4)\right)$$

$$u_t \Delta t + u_{tt}\frac{\Delta t^2}{2} + O(\Delta t^3) = -\lambda u_x \Delta x + \lambda u_{xx}\frac{\Delta x^2}{2} - \lambda u_{xxx}\frac{\Delta x^3}{6} + O(\Delta x^4)$$

$$\lambda = c \cdot \frac{\Delta t}{\Delta x}$$

$$u_t + c u_x = \frac{c\Delta x}{2} u_{xx} - \frac{c\Delta x^2}{6} u_{xxx} - u_{tt}\frac{\Delta t}{2}$$

$$+ O(\Delta t^2) + O(\Delta x^3)$$

$$u_t = -c u_x \Rightarrow u_{tt} = -c u_{xt}$$
$$= -\partial_x (u_t)$$
$$= c^2 u_{xx}$$

$$u_t + c u_x = \frac{c\Delta x}{2} u_{xx} - \frac{c^2 \Delta t}{2} u_{xx}$$
$$- \frac{c\Delta x^2}{6} u_{xxx} + O(\Delta t^2) + O(\Delta x^3)$$

$$= \frac{c\Delta x}{2}\left(1 - \frac{c\Delta t}{\Delta x}\right) u_{xx}$$
$$+ O(\Delta x^2)$$

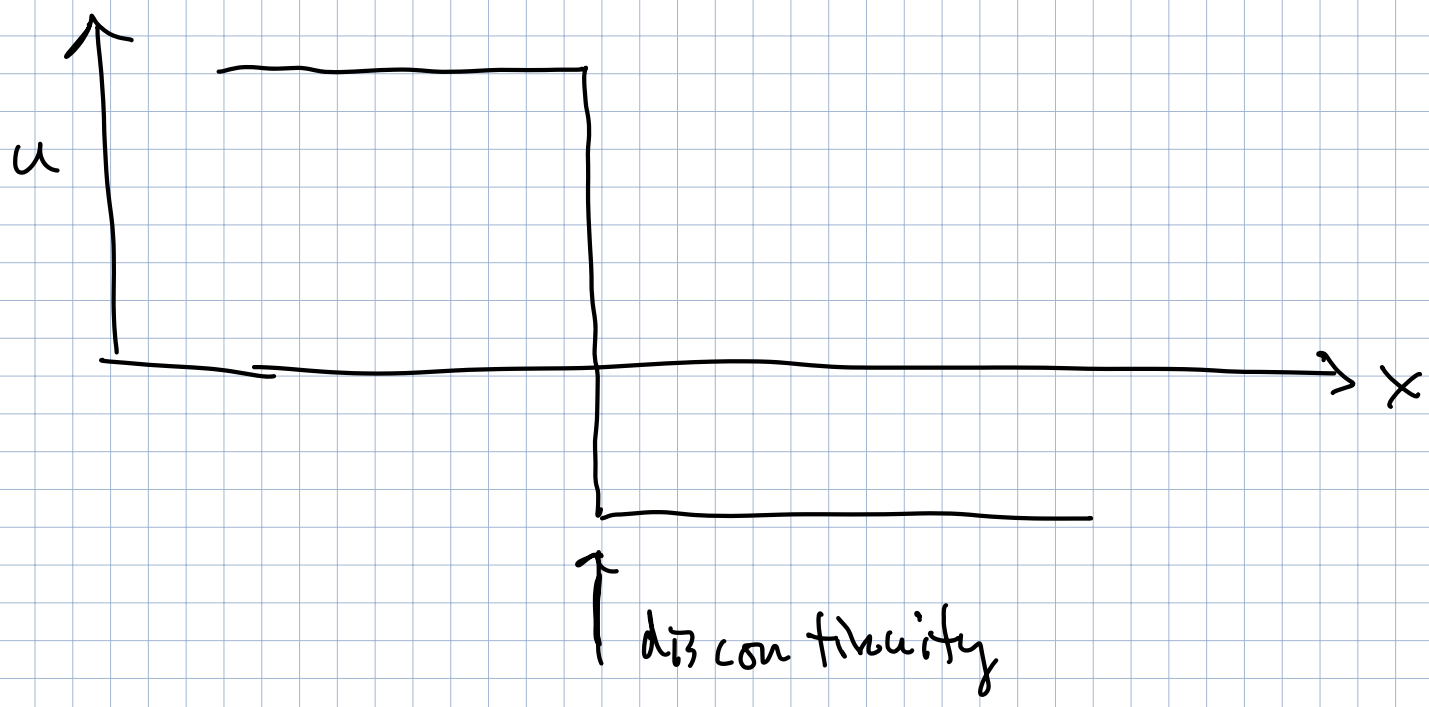
$$f_{k+1/2} = \text{flux}$$
$$= f^{uw}_{k+1/2} + \underbrace{\frac{c}{2}(1-\lambda)(u_{k+1\ell} - u_{k\ell})}_{\text{corrective flux}}$$

Lax-Wendroff:

$$u_{k\ell+1} = u_k - \frac{\lambda}{2}(u_{k+1} - u_{k-1\ell})$$
$$+ \frac{\lambda^2}{2}(u_{k+1\ell} - 2u_{k\ell} + u_{k-1\ell})$$

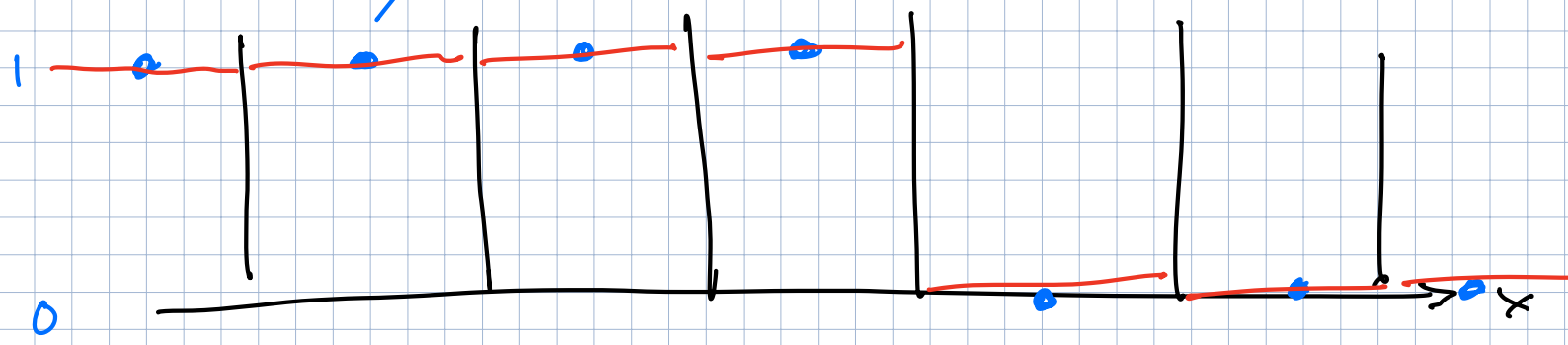
$$\rightarrow u_t + c u_x = -\frac{c\Delta x^3}{6}(1-\lambda^2) u_{xxx} + O(\Delta x^4)$$

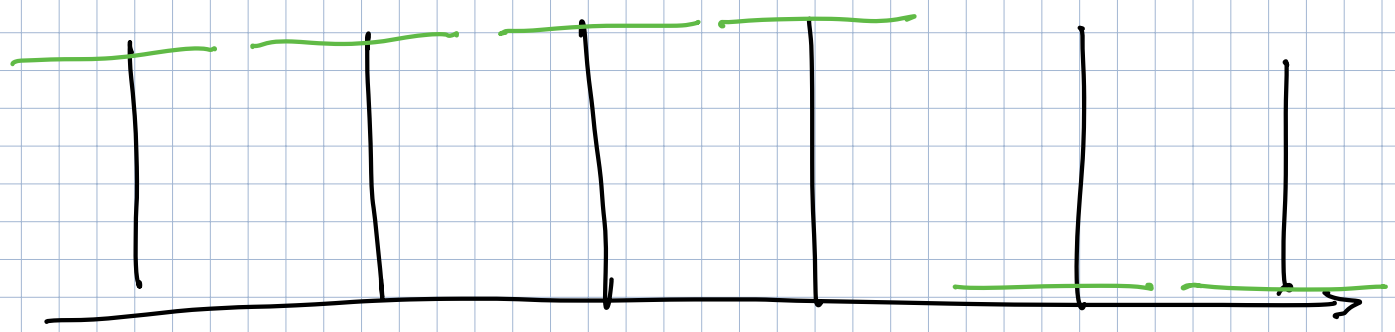
let $f_{k+1/2} = f^{uw}_{k+1/2} + N_\theta(u_{k+1,\ell}, u_{k,\ell}, u_{k-1,\ell})$

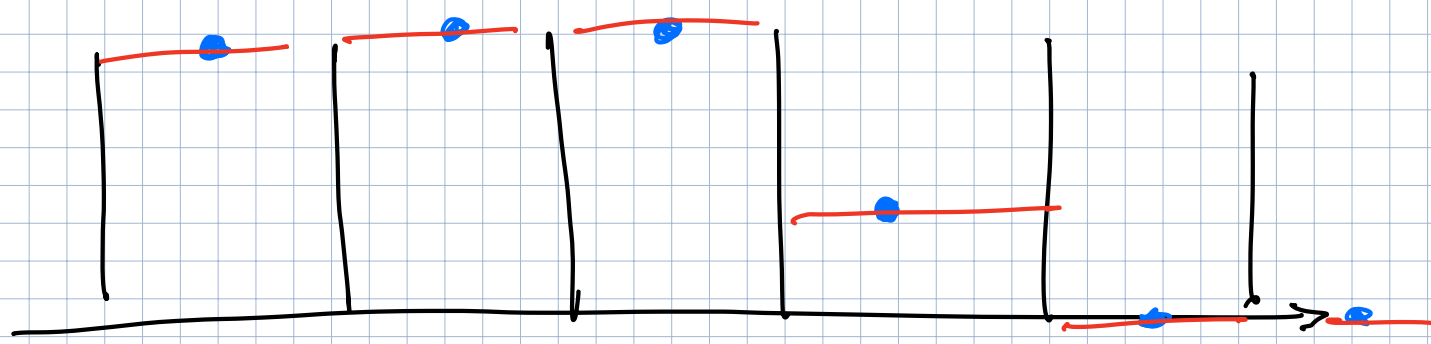


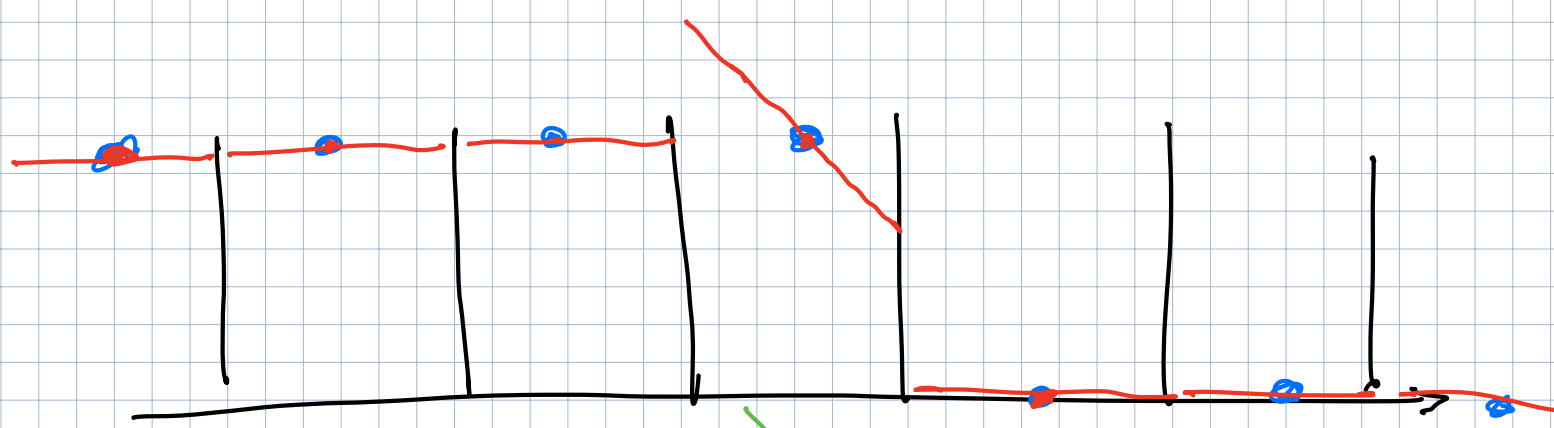
Godunov

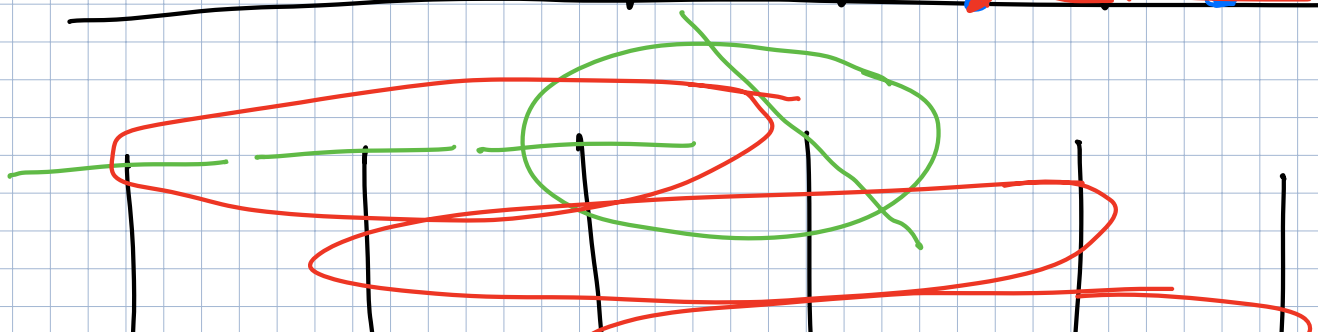
average or mean solution

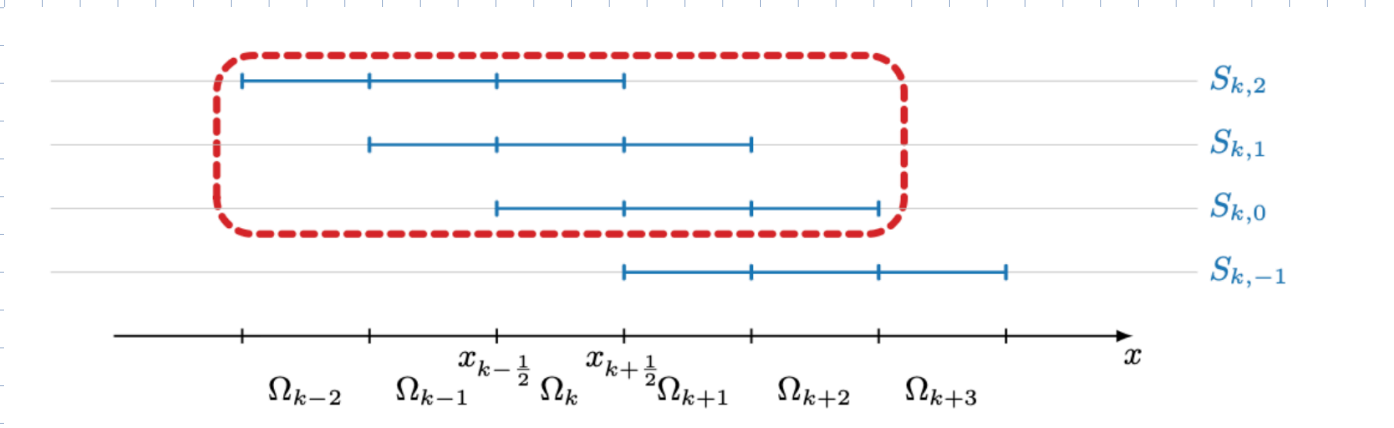












① Assume that the governing equations are ok.

$$\underline{u}_{\ell+1} = \mathcal{L}_c(\underline{u}_\ell) + N_\theta(\underline{u}_\ell)$$

coarse

Training targets correction to $\underline{u}_\ell^c$